APEX UPI – High-Concurrency PSP Switch & Ledger Engine

Document Control

Item	Details
Project Name	APEX UPI
Project Type	UPI Payment Processing Platform
Domain	FinTech / Digital Payments
Architecture	Event-Driven Microservices

1. Executive Summary

India's Unified Payments Interface (UPI) has transformed digital payments by enabling real-time, interoperable bank-to-bank transfers. As transaction volumes continue to grow exponentially, Payment Service Providers (PSPs) require highly available, fault-tolerant, and scalable systems capable of processing millions of transactions while ensuring consistency, auditability, and regulatory compliance.

Traditional monolithic payment systems face several limitations:

- Scalability bottlenecks
- Single points of failure
- Complex recovery procedures
- Limited observability
- Tight coupling between banking components
- Difficult reconciliation processes

The APEX UPI project aims to design and implement a distributed PSP Switch and Ledger Engine that simulates the core transaction processing capabilities of a UPI ecosystem. The solution leverages microservices, event-driven communication, ledger-based accounting, reconciliation workflows, and recovery mechanisms to provide a resilient payment processing platform.

The system processes payment requests, balance inquiries, VPA validations, transaction reversals, and reconciliation operations while maintaining complete audit trails and operational visibility.

2. Business Problem Statement

Current Challenges

Modern payment ecosystems face significant operational and technical challenges:

Challenge 1: Increasing Transaction Volume

UPI processes billions of transactions monthly. Existing architectures struggle to scale efficiently as transaction volumes increase.

Business Impact:

- Increased latency
- Transaction failures
- Customer dissatisfaction
- Operational overhead

Challenge 2: Service Failures

Payment transactions involve multiple systems:

- PSP
- NPCI
- Banking Systems
- Fraud Engines
- Audit Systems

Failure in any component can result in:

- Partial transaction completion
- Inconsistent balances
- Financial risk

Challenge 3: Lack of Recovery Mechanisms

When failures occur after debit but before credit:

- Funds may become temporarily unavailable
- Customers lose trust
- Manual intervention becomes necessary

Challenge 4: Poor Visibility

Operations teams often lack:

- End-to-end transaction tracking
- Real-time monitoring
- Root cause analysis capabilities

Challenge 5: Reconciliation Complexity

Transaction mismatches between:

- PSP Ledger
- NPCI Ledger
- Bank Ledger

can lead to:

- Financial discrepancies
- Compliance violations
- Operational losses

3. Business Opportunity

Scalability

Handle increasing transaction volumes through horizontal scaling.

Resilience

Continue processing despite component failures.

Auditability

Maintain complete transaction history.

Recoverability

Automatically resolve failed transactions.

Observability

Provide real-time transaction visibility.

Compliance

Maintain financial records and audit logs.

4. Project Vision

To build a highly available, fault-tolerant, event-driven UPI PSP Switch capable of processing payment transactions with enterprise-grade reliability, recovery mechanisms, ledger management, and operational observability.

5. Project Goals

Primary Goals

G1. Payment Processing

Enable secure real-time payment processing.

G2. Transaction Routing

Provide intelligent routing through NPCI simulation.

G3. Ledger Management

Maintain accurate financial records.

G4. Recovery Management

Automatically recover failed transactions.

G5. Reconciliation

Detecting and resolving transaction inconsistencies.

G6. Monitoring

Provide operational visibility through dashboards.

6. Business Objectives

Objective ID	Objective
BO-01	Process payment requests
BO-02	Validate customer requests
BO-03	Maintain accurate account balances
BO-04	Ensure transaction consistency
BO-05	Support transaction reversals
BO-06	Enable automated recovery
BO-07	Maintain audit trails
BO-08	Provide operational dashboards
BO-09	Support reconciliation processes
BO-10	Simulate real-world UPI workflows

7. Stakeholders

Primary Stakeholders

End Users

Customers performing UPI transactions.

Responsibilities:

- Initiate payments
- Check balances
- Track transaction status

Banks

Financial institutions maintaining customer accounts.

Responsibilities:

- Debit accounts
- Credit accounts
- Maintain balances

PSP Provider

Operates the payment switch.

Responsibilities:

- Transaction processing
- Validation
- Routing

NPCI

Central payment network.

Responsibilities:

- Transaction routing
- Settlement coordination

Secondary Stakeholders

Operations Team

Monitors platform health.

Audit Team

Reviews transaction records.

Compliance Team

Ensures regulatory adherence.

8. Scope

In Scope

Payments

- UPI payment initiation
- Payment processing
- Payment completion

Validation

- VPA validation
- Business rule validation
- Fraud checks

Banking Operations

- Debit transactions
- Credit transactions
- Balance verification

Ledger Management

- PSP ledger
- NPCI ledger

Recovery

- Timeout recovery
- Reversal processing

Reconciliation

- Ledger comparison
- Exception identification

Monitoring

- Transaction monitoring
- Service monitoring

Auditing

- Event tracking
- Historical analysis

9. Out of Scope

The following are intentionally excluded:

Production NPCI Integration

The system uses simulated NPCI interactions.

Real Banking Integration

CBS services are simulated.

Actual Settlement Processing

Inter-bank settlements are not implemented.

Mobile Applications

Only backend processing components are included.

KYC Management

Customer onboarding is excluded.

Merchant Settlement

Merchant settlement workflows are not included.

10. Business Assumptions

The project assumes:

A1

Every transaction possesses a unique Transaction ID.

A2

Every VPA is unique.

A3

Banks respond within acceptable timeout limits.

A4

Kafka provides reliable event delivery.

A5

PostgreSQL remains available during operations.

A6

Redis is used for caching and temporary state management.

A7

Services communicate over secured internal networks.

11. Business Constraints

Budget Constraint

The solution must utilize open-source technologies.

Infrastructure Constraint

Deployment is limited to five virtual machines.

Academic Constraint

The project serves as a simulation of actual UPI infrastructure.

Time Constraint

The platform must demonstrate end-to-end transaction processing within project timelines.

12. Success Criteria

The project shall be considered successful if:

Criteria	Target
Payment Processing	Successful completion
Balance Inquiry	Accurate response
Debit Accuracy	100%
Credit Accuracy	100%
Reversal Processing	Successful execution
Ledger Consistency	Maintained
Audit Logging	Complete
Recovery Handling	Operational
Dashboard Monitoring	Functional

13. Business Risks

Risk	Impact
Kafka Failure	Event delays
Database Failure	Data unavailability
CBS Failure	Transaction failures
NPCI Timeout	Pending transactions

Service Crash	Processing interruption
Network Failure	Communication disruption

14. Risk Mitigation Strategy

Kafka Retry Mechanism

Automatic retries and recovery.

Ledger-Based Recovery

Transaction state reconstruction.

Reversal Workflow

Compensation for failed transactions.

Audit Trails

Complete transaction tracking.

Reconciliation Engine

Mismatch detection and correction.

15. Expected Business Benefits

Operational Benefits

- Reduced transaction failure impact
- Faster recovery
- Better monitoring

Technical Benefits

- High scalability
- Loose coupling
- Improved maintainability

Educational Benefits

- Demonstrates enterprise payment architecture
- Simulates real-world PSP operations
- Provides hands-on exposure to event-driven systems

16. Conclusion

APEX UPI addresses the challenges of modern digital payment systems by introducing a scalable, fault-tolerant, and event-driven PSP Switch architecture. Through ledger-based accounting, automated recovery, reconciliation workflows, and comprehensive monitoring capabilities, the platform demonstrates how large-scale payment infrastructures can achieve reliability, transparency, and operational excellence while processing high-volume financial transactions.